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599 Edge AI and Robotics
Module 3 Video Redaction

Overview:

The purpose of this lab is to demonstrate facial redaction on a video file by drawing a rectangle around the face and obscuring it. This lab is directly related to the previous lab regarding image redaction, just applied to many images (frames) of a video. The lab compares performing the redaction on both the CPU and GPU.

CPU Implementation:

The main difference between the image and video redaction is that the video is read frame by frame. The instructions for the CPU remain largely the same, the haarcascade is still used to identify faces within a frame, cv2.rectangle is used to create the bounding box, and cv2.GaussianBlur is used to redact the face. It is just wrapped in a while loop until no more frames are to be read.

A video of the CPU result is visible under the media folder.

GPU Implementation:

Again, similar to the image redaction, the GPU implementation is virtually the same, frames are treated as a GpuMat() and CUDA is enabled for the classifier. Additionally inside the while loop for reading frames, the image must be sent and received back from the GPU.

A video of the GPU result is visible under the media folder.